A 2x2-Tile RV32EC Microcontroller SoC for Tiny Tapeout SKY 26e: Design, Verification and Physical Implementation in SkyWater 130 nm

Design: `tt_um_santhosh_rvsoc` | **Shuttle:** Tiny Tapeout SKY 26e | **Process:** SkyWater SKY130 (sky130A, 1.8 V, `sky130_fd_sc_hd`) | **Area:** 2x2 Tiny Tapeout tiles (334.88 x 225.76 um die) | **Target clock:** 64 MHz | **Author:** Santhosh (2026), based on TinyQV by Michael Bell

Status (10 Nov 2026): hardened (iteration 8); Magic DRC 0, KLayout DRC 0, LVS clean, antenna 0, hold met at all 9 corners, setup met at the typical corners (+1.59 ns); Tiny Tapeout precheck passed; RTL tests 11/11 and gate-level tests 11/11 pass. PYNQ-Z2 bitstream built with Vivado 2025.1 (timing met at 64 MHz) and verified in simulation on the RTL and on the post-route FPGA netlist. Not yet done: GitHub push and submission (manual), FPGA board test (board not yet available). A practical guide to using the chip is `docs/USER_GUIDE.md`.

> This report documents the complete design: what was built, why each decision was made, how it was verified, and how the layout was obtained, with enough detail to reproduce every result from the repository. Section 12 lists the exact commands and tool versions. Final physical numbers are also kept machine-generated in `reports/summary.md`; every hardening attempt is in `reports/sweep.md`.

Abstract

We present a complete RISC-V microcontroller system-on-chip that fits in four Tiny Tapeout tiles (about 0.073 mm2 of core area) of the SkyWater 130 nm process. The SoC is built around TinyQV, an RV32EC core that processes data four bits per clock cycle, executes code in place from an external QSPI flash and uses two external QSPI PSRAMs as data memory, so it needs no on-chip SRAM. Around the core we integrated a UART with a programmable baud-rate divisor, eight general-purpose inputs and outputs, two edge-triggered external interrupts and a 32-bit timer with a compare interrupt. The timer is implemented nibble-serially, in step with the CPU's 4-bit datapath, which makes torn 32-bit reads impossible and saves area. Verification combines a firmware self-test that runs every supported instruction, pin-level system tests and random instruction tests against a reference model, all repeated on the gate-level netlist, plus a cycle-accurate FPGA harness for a PYNQ-Z2 board, built with Vivado (timing met at 64 MHz) and verified down to its post-route netlist. The self-test found two defects in the upstream core (a hang on `fence` and wrong operands for the CSR immediate instructions); both are fixed. Physical implementation used the official Tiny Tapeout flow (LibreLane 3.0.14). Fitting the design into 2x2 tiles took several iterations, which are documented: removing logic outside the specification, a serial timer and narrower peripheral read bus, limiting hold repair to real violations and finally a higher placement density reduced global-routing demand from 98.6 % to 64.8 % of capacity and made the design routable. The final layout passes DRC, LVS, antenna and hold checks at all corners, meets setup at 64 MHz at the typical corners (+1.59 ns slack), and passes the Tiny Tapeout precheck.

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1. Introduction

1.1 Motivation

Tiny Tapeout (TT) lets individual designers put digital designs on a real chip by sharing one multi-project die. Each design gets one or more "tiles" of roughly 161 x 112 um and 26 signals: 8 inputs, 8 outputs, 8 bidirectional pins, a clock and a reset. A processor capable of running compiled C code is far larger than a single tile, and on-chip memory is very expensive in area. TinyQV (Michael Bell, TT06) showed that a 32-bit RISC-V CPU fits in 2x2 tiles if (a) the datapath is 4 bits wide and processes a 32-bit word over 8 clock cycles, and (b) all memory is external, reached over a 4-bit QSPI bus.

1.2 Goal and constraints

The goal was a submission-ready TinyQV-class microcontroller for the SKY 26e shuttle (projects close 30 Nov 2026; first planned for SKY 26d, Section 2.1) with:

- RV32E + C instruction set, execute-in-place from QSPI flash, PSRAM as RAM;
- UART (8N1, programmable baud), 8 GPIO out, 8 GPIO in, 32-bit timer with compare interrupt, external interrupt pins;
- 64 MHz clock, timing closed in the TT flow's STA;
- 2x2 tiles (fallback 3x2 only after an iteration budget), clean DRC/LVS/antenna;
- RTL and gate-level tests, demo firmware, documentation and one-command automation.

Free tiles are scarce, so area had priority over features.

1.3 Contributions

1. An integrated SoC (`src/project.v`) that keeps the TT06 TinyQV memory map and pinout but replaces debug/SPI blocks with the MVP peripherals of the spec.
2. A nibble-serial 32-bit timer (`src/timer.v`) locked to the CPU's nibble counter.
3. Two upstream core fixes found by our own ISA self-test (Section 5.2).
4. A layered verification environment that runs the same tests on RTL and gates, plus an FPGA harness that emulates the QSPI Pmod in block RAM.
5. A documented area/routability closure study in 2x2 tiles (Section 8).

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2. Background

2.1 Tiny Tapeout SKY 26e

The project was developed for SKY 26d. On 10 Nov 2026 that shuttle had only 3 free tiles, too few for a 2x2 design, so it moved to SKY 26e, which closes on the same day (30 Nov 2026) with the same process. The two shuttles use the same build action (`tt-gds-action@ttsky26d` and `@ttsky26e` are one commit, with LibreLane 3.0.14), so nothing in the design or its results changes.

Item	Value
Process	SkyWater SKY130, PDK variant sky130A (open_pdks 8afc8346)
Std cells	`sky130_fd_sc_hd` (high density)
Supply	1.8 V core

Item	Value
Tile	about 161 x 112 um; 2x2 gives a 334.88 x 225.76 um die, 329.36 x 220.32 um core
Routing layers for user designs	met1 to met4 (met5 is reserved for the TT power grid)
Flow	tt-support-tools + LibreLane 3.0.14 (OpenROAD, Yosys, Magic, KLayout, Netgen)
I/O	8 in (`ui_in`), 8 out (`uo_out`), 8 bidir (`uio_*` + `uio_oe`), `clk`, `rst_n`, `ena`
I/O speed	outputs about 33 MHz, inputs about 66 MHz through the TT multiplexer

2.2 TinyQV

TinyQV (Apache-2.0) is an RV32EC CPU designed for minimal area. Its 32-bit registers are shift registers that rotate by 4 bits every clock, so one 4-bit ALU processes a 32-bit operand in 8 clocks. A QSPI memory controller fetches 16-bit instruction parcels from flash in continuous-read (XIP) mode and performs data reads and writes to two QPI PSRAMs. Upstream TT06 also had an SPI controller, a debug UART and debug outputs. We started from the TT06 version (`tt06-tinyQV`, tinyQV commit dfbd5e38), not the later multi-tile "crowd-sourced" TinyQV.

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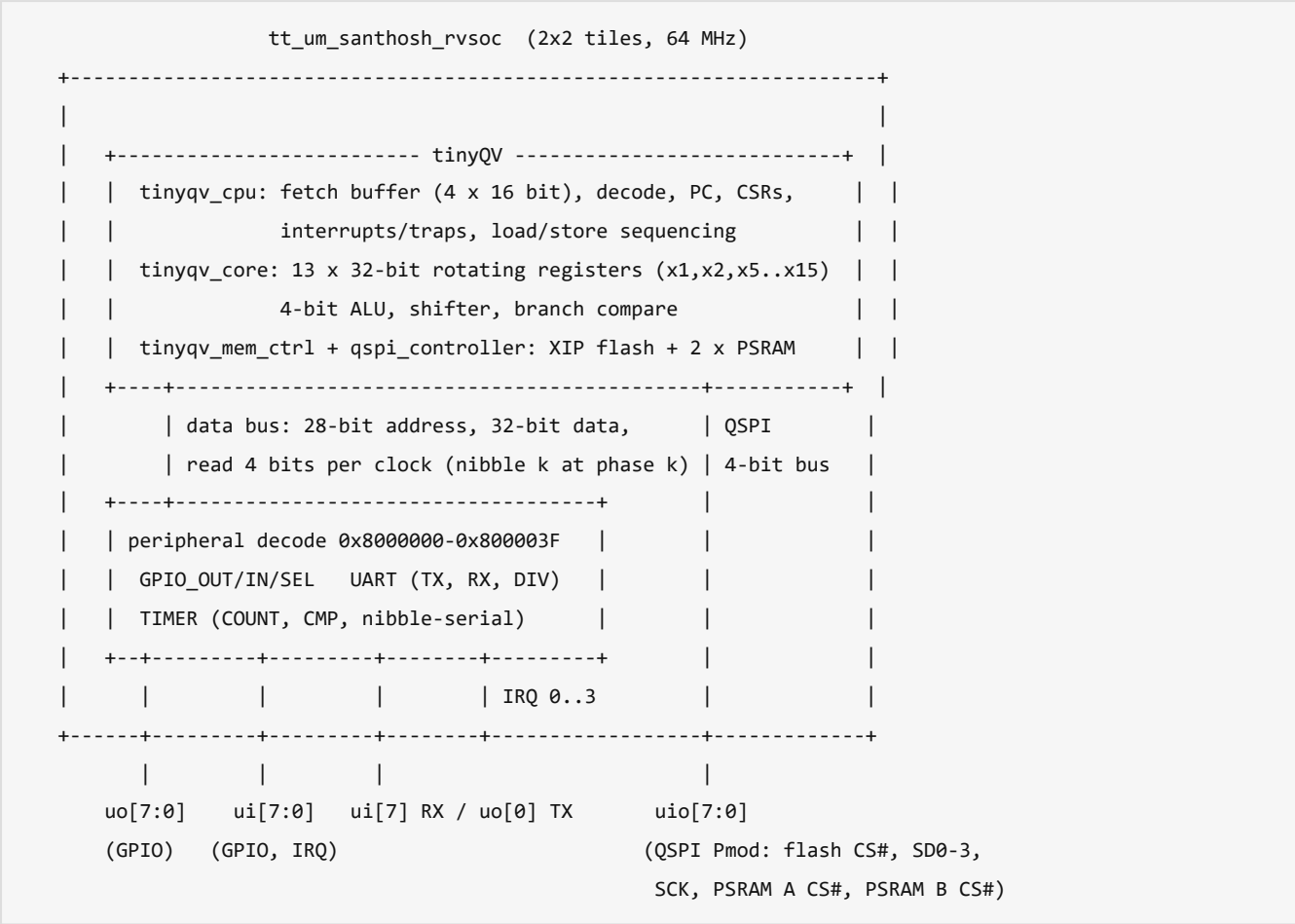
3. Requirements and how they are met

#	Requirement	Where met
R1	TT top-level interface, 26 signals, no floating outputs	`src/project.v` (all outputs driven)
R2	32-bit RISC-V, 4-bit datapath, RV32E + C + minimal Zicsr/trap	TinyQV core, Sections 4.2 and 5
R3	Execute in place from QSPI flash, PSRAM A/B as RAM, no on-chip SRAM	Section 4.3
R4	UART 8N1 with programmable divisor	`UART_DIV` register, Section 4.4
R5	GPIO: 8 out (read/write), 8 in	`GPIO_OUT`, `GPIO_IN`, `GPIO_SEL`
R6	32-bit timer with compare interrupt	`src/timer.v`, Section 4.5
R7	Keep upstream memory map and register layout	Section 4.6
R8	64 MHz, setup/hold pass in the flow's STA	Section 8.5
R9	2x2 tiles, no DRC/LVS/antenna errors	Section 8
R10	Tests: boot, UART "Hello, TT!", GPIO loopback, timer IRQ, PSRAM, every instruction; RTL and gate level	Section 7
R11	C firmware for RV32E with banner, GPIO pattern, timer tick	`firmware/`, Section 6
R12	Datasheet, reports, one-command automation	`docs/`, `reports/`, `scripts/run_all.sh`

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4. Architecture

4.1 Block diagram



4.2 CPU core

Datapath. Every 32-bit quantity (register, ALU operand, address, store data, CSR) moves 4 bits per clock. A 3-bit nibble counter (`counter_hi`) runs 0..7 continuously; in phase *k* the core works on bits [4k+3:4k]. A simple instruction therefore takes at least 8 clocks; shifts, compares and loads/stores take more.

Register file. RV32E has 16 registers. `x0` is constant 0, and TinyQV hard-wires `gp` (x3) = 0x1000400 and `tp` (x4) = 0x8000000 (the PSRAM global pointer area and the peripheral base), so only 13 registers are stored: 13 x 32 = 416 flip-flops, each register rotating by 4 bits per clock. Upstream places a `sky130_fd_sc_hd__dlygate4sd3_1` delay cell on each rotating bit (364 cells) as a fixed hold-time fix (see Section 8.3, iteration 5).

Instruction fetch. The CPU keeps a small prefetch buffer of 16-bit parcels filled by the QSPI controller from flash in continuous-read mode. Branches restart the fetch. Code may only execute from flash (24-bit code addresses).

ISA. RV32E base, C (compressed), Zicsr, `ecall`, `ebreak`, `mret`. TinyQV also decodes some Zcb compressed instructions and custom multi-register loads/stores; they are kept (removing them saves only about 0.8 % area). No misaligned-access or illegal-instruction trap is implemented (upstream behaviour).

CSRs. `mstatus` (MIE, MPIE, plus TinyQV's non-standard trap-enable bit), `misa` (reads 0x40000014 = RV32 + C + E), `mie`, `mip`, `mepc` (24 bits), `mcause`. The Zicntr counters were removed for area (Section 5).

Traps and interrupts. Fixed vectors: reset 0x0, trap (`ecall`, `ebreak`) 0x4, interrupt 0x8. Four interrupt sources map to `mip`/`mie` bits 16..19, with `mcause` = 0x80000010 + n:

n	Source	Type
0	`ui_in[0]`	rising edge, latched in `mip`, cleared by software
1	`ui_in[1]`	rising edge, latched in `mip`, cleared by software
2	UART byte received	level, cleared by reading `UART_DATA`
3	Timer `COUNT == CMP`	level, cleared by writing `TIMER_CMP`

4.3 Memory subsystem

All memory is external on one 4-bit QSPI bus (the Tiny Tapeout QSPI Pmod: 16 MB W25Q128-class flash and two 8 MB APS6404-class PSRAMs):

Address	Device	Access
0x0000000-0x0FFFFFF	flash, continuous-read mode (`EBh` with mode bits `A0`)	execute, read
0x1000000-0x17FFFF	PSRAM A, QPI mode (`0Bh` read with 4 dummy clocks, `02h` write)	read, write
0x1800000-0x1FFFFFF	PSRAM B, QPI mode	read, write
0x8000000-0x800003F	peripheral registers	read, write

SCK runs at clk/2 (32 MHz at 64 MHz). Read data come back through the TT I/O multiplexer and pads with a board-dependent delay, so the controller has a programmable read latency (0..5 extra half SCK cycles). The latency is sampled from SD2:SD0 (`uio[4]`, `uio[2]`, `uio[1]`) while `rst_n` is low. The flash must already be in continuous-read mode and the PSRAMs in QPI mode before reset is released; on the TT demo board the RP2040 does this.

4.4 Peripherals

Peripheral bus. Addresses 0x8000000-0x800003F with `addr[1:0] == 0` select a 32-bit register by `addr[5:2]`; all accesses complete in one cycle. Writes arrive with the whole 32-bit store data valid and always in nibble phase 0 (verified by a simulation assertion). The CPU reads `data_in[4k+3:4k]` in phase k, and the sign bit of byte/halfword loads is taken from the current nibble too. So the SoC selects only the current 4-bit nibble and replicates it on the 32-bit bus: a 4-bit-wide read multiplexer instead of a 32-bit one.

GPIO. `GPIO_OUT` drives `uo_out[7:2]` directly and `uo_out[1:0]` when the corresponding `GPIO_SEL` bit is set (otherwise these are UART TX and RTS). Reading `GPIO_OUT` returns the actual output pins. `GPIO_IN` returns `ui_in[7:0]`.

UART. Ben Marshall's 8N1 UART (MIT), as modified by Michael Bell, now takes its bit period from the 13-bit `UART_DIV` register (clocks per bit - 1) instead of a synthesis parameter. Reset value 555 gives 115,108 baud at 64 MHz (-0.08 %). The receiver has a one-byte buffer and an active-low RTS output.

4.5 Nibble-serial timer

A parallel 32-bit timer (counter, compare register, 32-bit incrementer, 32-bit comparator, 32-bit read mux) cost about 3,200 um2. Because TinyQV consumes data 4 bits per clock, the timer can be serial as well:

- `COUNT` and `CMP` are 32-bit registers that rotate right by 4 bits every clock, locked to the CPU's nibble counter (`nib_phase`), so nibble *k* is in bits [3:0] exactly when the CPU reads nibble *k*;
- a 4-bit adder with a carry flip-flop increments `COUNT` by 1 as it rotates, once per 8 clocks (8 MHz at 64 MHz). All nibbles of one read belong to the same value, so **a 32-bit read can never be torn**;
- a 4-bit comparator accumulates `COUNT == CMP` over one rotation and sets the sticky IRQ flag at phase 7;
- writes (always in phase 0) load the value pre-rotated by one nibble and suppress the increment and compare for that pass.

Result: 2,332 um2 and no width-dependent carry chain.

4.6 Register map

Address	Name	Access	Description
0x8000000	GPIO_OUT	R/W	output value; read returns `uo_out`
0x8000004	GPIO_IN	R	`ui_in`
0x800000C	GPIO_SEL	R/W	bit 0/1: `uo[0]`/`uo[1]` from GPIO_OUT instead of UART TX/RTS; bits 7:2 read 1
0x8000010	UART_DATA	R/W	write: send byte; read: received byte
0x8000014	UART_STATUS	R	bit 0 TX busy, bit 1 RX valid
0x8000028	TIMER_COUNT	R/W	32-bit, +1 per 8 clocks
0x800002C	TIMER_CMP	R/W	IRQ 3 while COUNT == CMP; writing clears IRQ
0x8000034	UART_DIV	R/W	clocks per bit - 1 (13 bits), reset 555
others	-	R	read 0xFFFFFFFF

GPIO and UART addresses are identical to TT06 TinyQV. The new registers use addresses that were unused upstream.

4.7 Pinout

Pin	Function		Pin	Function
ui[0]	IRQ0 / GPIO in0		uo[0]	UART TX (or GPIO out0)
ui[1]	IRQ1 / GPIO in1		uo[1]	UART RTS (or GPIO out1)
ui[2..6]	GPIO in2..6 (demo: ui[3:2] mode, ui[4] sim-fast)		uo[2..7]	GPIO out2..7
ui[7]	UART RX / GPIO in7			
uio[0]	flash CS#		uio[4]	SD2
uio[1]	SD0		uio[5]	SD3
uio[2]	SD1		uio[6]	PSRAM A CS#
uio[3]	SCK		uio[7]	PSRAM B CS#

`uio_oe` is 0 while `rst_n` is low; after reset CS#/SCK are outputs and SD0-3 are driven only during command/address/write phases. The pin-by-pin user guide is `docs/IO_GUIDE.md`.

4.8 Reset and boot sequence

`rst_n` is registered on the falling clock edge (`rst_reg_n`), and the CPU re-registers it on the rising edge. Boot procedure:

1. hold `rst_n` low (all `uio` inputs); drive SD2:SD0 = read latency; clock >= 8 times;
2. release the QSPI lines and raise `rst_n` while `clk` is high;

3. the CPU fetches from flash address 0, which holds `j reset_handler`.

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5. Changes relative to upstream TinyQV (TT06)

Change	File	Reason	Effect
SPI controller, debug UART, debug muxes removed	`project.v`	not in the spec; area	- about 2,000 um2
UART bit period from `UART_DIV` register	`uart_tx.v`, `uart_rx.v`, `project.v`	spec R4	+13-bit register
32-bit timer + compare IRQ (nibble-serial)	`timer.v`	spec R6	2,332 um2
IRQ 3 = timer (upstream: UART TX idle)	`project.v`	spec R6	-
`uo[7:2]` always GPIO; GPIO_SEL is 2 bits	`project.v`	simpler, area	-
4-bit peripheral read bus	`project.v`	area	small saving
`fence` **decoded as NOP**	`decode.v`	**bug: upstream stalled forever on** `fence`	RV32E base complete
CSR immediate forms use zimm	`core.v`	**bug: upstream used register rs1 instead of the immediate**	correct `csrrsi` / `csrrci` / `csrrwi` (zimm[4] reads 0)
custom 16-bit multiplier removed	`core.v`	non-standard, not used by compiled RV32EC code; area	- about 2,400 um2
Zicntr counters removed	`core.v`	not required; TIMER_COUNT provides time; routability	- about 1,600 um2
simulation model: memories read via continuous assigns	`test/sim_qspi.v`	Icarus elaboration time	test only

5.1 Why the bugs matter

- **FENCE hang.** The upstream decoder did not recognise the MISC-MEM opcode (`0001111`); the core never completed the instruction and stalled. GCC emits `fence` for `__sync_synchronize()` and some atomics; any such program would hang. On a single in-order hart without caches FENCE is correctly a NOP, so it is now decoded as `addi x0, x0, ...`.
- **CSR immediates.** For `csrrwi/csrrsi/csrrci` the operand must be the 5-bit immediate in the rs1 field. Upstream used the register number as a register read, so for example `csrsi mstatus, 8` read register x8 (s0) instead of the constant 8, which silently produced wrong interrupt-enable states. The fix uses the rs1 field (zimm[3:0]) as the operand. zimm[4] is not available because RV32E register numbers are 4 bits; it reads as 0, which can only matter for `csrrwi mepc` with bit 4 set.

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6. Firmware

`firmware/` (C + assembly, built with xPack `riscv-none-elf-gcc` 14.3, `-march=rv32ec_zicsr -mabi=ilp32e -Os`, about 4.6 KB):

- `start.S`: vectors at 0x0/0x4/0x8, stack in PSRAM A, `.data` copy, `.bss` clear, trap and interrupt entry saving the 10 caller-saved RV32E registers;
- `link.ld`: code and constants in flash at 0x0, data/bss/stack in the first 64 KB of PSRAM A; `__global_pointer$` = 0x1000400 to match the hard-wired `gp`;
- `isa_test.S`: self-checking test of every supported instruction (RV32E ALU register/immediate, LUI/AUIPC, all branches taken and not taken, JAL/JALR including bit-0 clearing, all loads/stores from PSRAM and flash, FENCE, x0 and hard-wired gp/tp, all six Zicsr forms on mstatus/mie/mepc/misa, ECALL/EBREAK with mcause/mepc, and every RV32EC compressed instruction);
- `main.c`: modes chosen by `ui[3:2]` at reset (00 demo, 01 GPIO mirror) and `ui[4]` (simulation fast mode: 2 Mbaud, 250 us tick). The demo prints `Hello, TT!`, runs the self-tests (ISA, PSRAM, TIMER, GPIO), then runs an LED chaser on `uo[7:2]` driven by a 100 ms timer interrupt, prints `tick N` every second, echoes UART input and prints `IRQ0`/`IRQ1` on external interrupts.

Measured cost of an interrupt with a C handler, running from QSPI flash with the stack in PSRAM: about 2,400 clocks (37 us at 64 MHz). This was found when a 300-tick (2,400-clock) self-test period livelocked the CPU (Section 7.5).

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7. Verification

7.1 Strategy

Level	What	Tool	Runs on
1	Firmware self-test (ISA, PSRAM, timer, GPIO) inside the chip	RISC-V firmware	RTL, gates, FPGA, silicon
2	Pin-level system tests (`test_soc.py`)	cocotb 2.1 + Icarus 11	RTL, gates
3	Random instruction tests vs. reference model (`test_isa_random.py`, upstream)	cocotb + riscv-model	RTL, gates (fewer seeds)
4	FPGA harness simulation (`fpga/sim/tb_fpga.v`, `tb_fpga_board.v`)	Icarus, Vivado xsim	RTL + FPGA emulator; Vivado post-route netlist
5	Static: lint (Verilator), STA (OpenSTA, 9 corners), DRC/LVS/antenna, TT precheck	LibreLane, Magic, KLayout, Netgen	layout

7.2 Testbench

`test/tb.v` instantiates the chip with:

- **QSPI Pmod model** (`sim_qspi.v`, upstream): 64 KB flash loaded with the firmware hex, two 64 KB PSRAMs, and a programmable read-latency pipeline (`latency_cfg`), also driven onto SD2:SD0 during reset;
- a switch (`use_model`) that lets Python drive the QSPI bus directly for the upstream instruction-level tests;
- a **Verilog UART monitor** on `uo[0]` (programmable bit period) that hands bytes to Python, so Python never samples the pin itself (fast, works on gates).

7.3 Test list

Test	Checks	Requirement
`test_boot_hello`	reset + latency config; boot from flash; `Hello, TT!` decoded from `uo[0]` at 115200 (latency 1) and 2 Mbaud (latency 2 and 3)	R3, R10
`test_isa_selftest`	firmware ISA self-test prints `ISA PASS`	R2, R10
`test_psram`	`PSRAM PASS` and the written words found inside both PSRAM models	R3, R10
`test_timer_irq`	`TIMER PASS` (3 interrupts, mcause 0x80000013), then the interrupt-driven LED chaser stepping every 16,000 +/- 500 clocks on the pins, and `tick 10`	R6, R10
`test_gpio_loopback`	`GPIO PASS` (write/read-back incl. GPIO_SEL), then mirror mode: 12 random `ui_in` patterns appear on `uo[7:2]`	R5, R10
`test_uart_echo`	characters sent into `ui[7]` are echoed (UART RX path, polling)	R4
`test_ext_irq`	rising edges on `ui[0]`, `ui[1]` print `IRQ0`, `IRQ1`	R2 interrupts
`test_start` (upstream)	UART TX/RX, GPIO via Python-driven instructions	R4, R5
`test_load_bug` (upstream)	regression for a TinyQV load hazard	R2
`test_random_alu` (upstream)	25 seeds x 200 random ALU/compressed instructions vs. riscv-model	R2
`test_random` (upstream)	5 seeds x 1000 random instructions incl. loads/stores to PSRAM	R2

7.4 Results

RTL: Icarus Verilog 11, cocotb 2.1, `test/` with `make` (results in `logs/rtl_results.xml`). Gate level: the powered post-route netlist `runs/wokwi/final/pnl/tt_um_santhosh_rvsoc.pnl.v` with the sky130 cell models, `scripts/gl_test.sh` (results in `logs/gl_results.xml`). Wall-clock times are for this laptop (WSL2, 8 threads).

Test	RTL	RTL time (s)	Gate level	GL time (s)
`test_boot_hello`	PASS	94	PASS	191
`test_isa_selftest`	PASS	24	PASS	74
`test_psram`	PASS	42	PASS	128
`test_timer_irq`	PASS	241	PASS	1,699
`test_gpio_loopback`	PASS	20	PASS	64
`test_uart_echo`	PASS	123	PASS	393
`test_ext_irq`	PASS	181	PASS	292
`test_start`	PASS	140	PASS	158
`test_load_bug`	PASS	1	PASS	1
`test_random_alu`	PASS (25 seeds)	765	PASS (3 seeds)	226
`test_random`	PASS (5 seeds)	1011	PASS (1 seed)	112
Total	**11/11**	**2,642**	**11/11**	**3,339**

On gates, all nine deterministic tests (the seven `test_soc` tests, `test_start` and `test_load_bug`) finish at exactly the same simulated time as on RTL (for example 1,381,630.32 ns for `test_boot_hello`): the post-route netlist reproduces the RTL behaviour cycle for cycle, including every UART bit. The random-instruction tests use fewer seeds on gates (3 and 1 instead of 25 and 5) to keep the run short; the whole gate-level suite takes 56 minutes here, well inside the 6-hour limit of the GitHub `gl_test` job.

The FPGA harness simulation (`make -C fpga sim`) also passes: banner, all four self-tests and the LED chaser, with the BRAM flash/PSRAM emulator at read latency 2. The Vivado runs of the FPGA build (RTL and

post-route netlist) are in Section 9; a test-by-test explanation of all outputs is in `docs/USER_GUIDE.md` section 3.

7.5 Issues found by verification

Found by	Issue	Resolution
ISA self-test	CPU hangs at `fence`	decoder fix (Section 5.1)
ISA self-test	`csrrsi` / `csrrci` / `csrrwi` used a register	core fix (Section 5.1)
`test_gpio_loopback`	firmware switched `uo[0]` to GPIO while GPIO bit 0 was 0, causing a false UART start bit	firmware writes the value before selecting the pin
`test_timer_irq`	300-tick self-test period shorter than the ~2,400-clock ISR: livelock	2,000-tick period; documented ISR cost
`test_uart_echo`	2 Mbaud back-to-back bytes overrun the 1-byte RX buffer while polling	test types one character at a time; buffer documented
tooling	Icarus elaboration of the PSRAM model took minutes	continuous-assign read path
tooling	`buf #1` in TinyQV needs a 1 ns default timescale	`+timescale+1ns/1ps` for plain Icarus runs

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8. Physical implementation

8.1 Flow

Tiny Tapeout local hardening flow, exactly as in the official guide: `tt/tt_tool.py --pdk sky130A --create-user-config`, then `--harden`, which runs LibreLane 3.0.14 in Docker: Verilator lint, Yosys synthesis (`AREA 0`), OpenROAD floorplan/PDN/placement/CTS/resizer/global and detailed routing, antenna repair, OpenSTA multi-corner STA, Magic and KLayout streaming and DRC, Netgen LVS. Only the user section of `src/config.json` was changed.

8.2 Floorplan

Item	Value
Die	334.88 x 225.76 um (2x2 tiles)
Core	2.76..332.12 x 2.72..223.04 um, 72,565 um2
Routing	met1-met4 (met4 also carries the TT power stripes, pitch 38.87 um)
I/O pins	fixed by the TT template (top edge)
Rows	81 rows of `unithd` sites

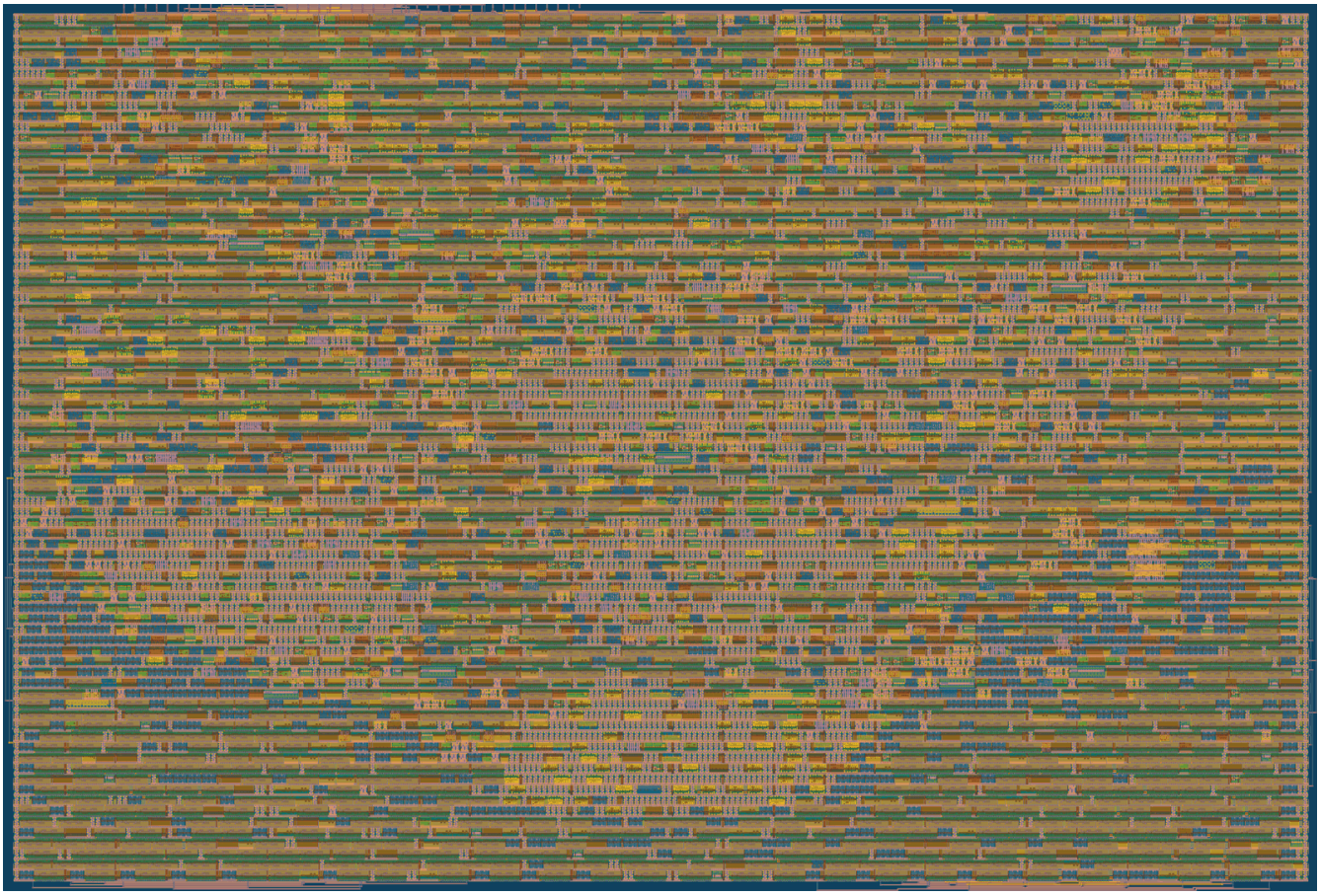


Figure 1. Final layout (2x2 tiles), rendered with `tt_tool --create-png` *(the same renderer as the Tiny Tapeout GitHub Action). Pins are on the top edge.* `images/layout_zoom.png` *shows the centre at full resolution.*

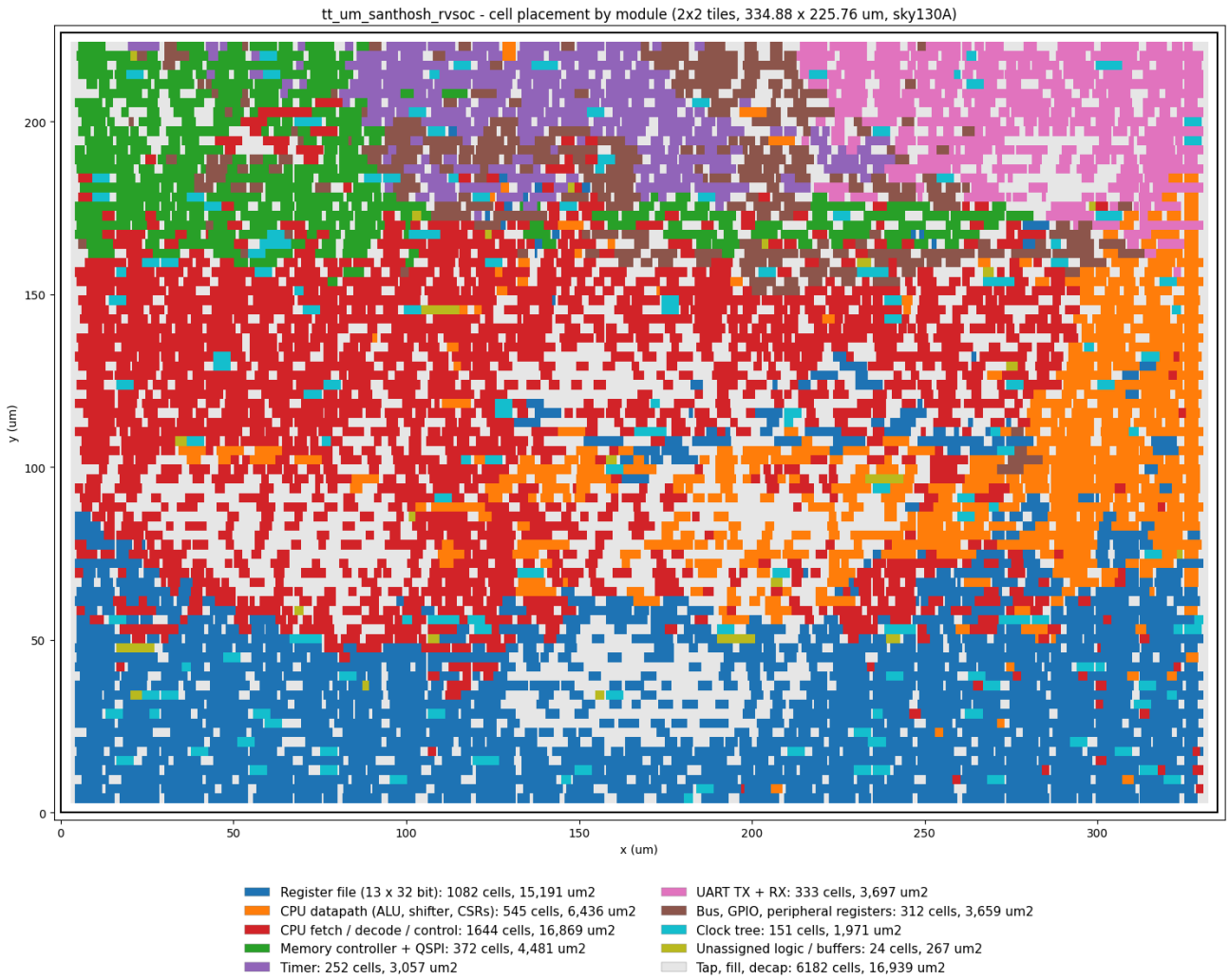


Figure 2. Where each block was placed (`scripts/docs/module_map.sh`). The netlist is flat after synthesis, so a cell is labelled from the hierarchical name of the net it drives, and cells driving unnamed nets take the majority label of their neighbours; the split between "fetch/decode" and "datapath" is therefore approximate. The placer put the register file along the bottom edge, the QSPI controller next to the `uio` pins (top left), the UART at the top right and the timer and peripheral registers in between.

8.3 Closure iterations

`reports/sweep.md` lists every attempt. Summary:

Iter	Change	Result	Lesson
it1	TT defaults, 64 MHz, density 60	detailed placement failed after post-CTS repair	the resizer tried to fix setup at the slow corner (ss, 100 C, 1.60 V), where the 64 MHz CPU path fails by ~7 ns; buffering overfilled the core
it2	repair corners = typical + fast (the flow only enforces setup at typical); density 80	setup met (+3.5 ns at tt); placement failed again	0.1 ns hold margin made the resizer buffer ~460 flop-to-flop paths of the rotating registers
it3	removed custom multiplier; serial timer; hold margin 0.05/0.03	placed, CTS'd, timing met; global routing 98.6 % full, detailed routing did not converge	area alone was not enough; routing demand is the limit
it4	removed Zicntr; 4-bit read bus; max fanout 16	routing 85.0 % full; detailed routing plateaued at ~7,400 violations	still too much wiring

Iter	Change	Result	Lesson
it5	removed the 364 fixed register delay cells	resizer added 641 hold buffers instead; no gain	upstream's fixed delays are the efficient hold fix; reverted
it6	**hold-repair margin 0.0 post-CTS / 0.02 post-GRT**	only **3 hold buffers** instead of 290; routing **68.9 %** full; detailed routing fell to 2-3 violations but stayed there for 50+ passes	margin-driven hold buffers were the dominant routing load
it7	density 77	identical to it6 (placer clamps to 0.84)	density below 84 % has no effect for this design
it8	**it6 + density 86**	**routing 64.8 % full (165 overflows); 0 violations after 20 passes; all signoff checks pass**	**final design**

The key insight is that in a nibble-serial CPU almost every flip-flop feeds another flip-flop directly. With a hold margin, hundreds of such paths sit just below the margin, and buffering them all adds both area and long detours. Fixing only real violations (hold >= 0 is still enforced at every corner at signoff) cut routing demand from 85.0 % to 68.9 %. The last step was the placement density: the global placer never spreads this design below 84 %, and asking for 86 % kept connected cells closer, shortened the estimated wiring and removed the last congestion spots, so detailed routing reached zero violations.

Iterations 5-8 were run in parallel in scratch copies (`.work/itN`) to save time; the RTL and configuration of it8 are identical to the files in `src/`, and `runs/wokwi` is the it8 run.

8.4 Timing methodology

The flow analyses 9 corners (nominal/min/max RC x tt/ss/ff). Following the TT flow defaults, setup is enforced at the typical corners (`TIMING_VIOLATION_CORNERS = *tt*`) and hold at all corners. Repair is run at the typical and fast corners (`RSZ_CORNERS`). The slow-corner (ss, 100 C, 1.60 V) setup slack is reported for information: the 64 MHz CPU path does not close there (worst slack -12.17 ns), which means at 100 C and 1.60 V the chip would need a period of about 28 ns (about 36 MHz). Tiny Tapeout chips run at room temperature on 1.8 V, where the typical corners apply.

The flow also reports maximum slew and capacitance, but does not treat them as failures (`MAX_SLEW_VIOLATION_CORNERS` and `MAX_CAP_VIOLATION_CORNERS` are empty in the TT configuration). See Section 8.5 for what they show here.

8.5 Final results

All numbers are from `runs/wokwi/final/metrics.json` and the post-route STA in `runs/wokwi/55-openroad-stapostpnr/` (also in `reports/summary.md` and `reports/sweep.md`).

Metric	Value
Standard cells	5,752 (2,870 combinational, 986 flip-flops, 169 clock tree, 492 buffers/inverters, 132 timing-repair buffers, 1,037 tap, 66 antenna diodes)
Cell area	56,923 um2 of 72,565 um2 core: **78.4 % utilisation**; the rest is 5,145 fill/decap cells
Hold buffers	0 (the 364 fixed register-file delay cells handle the rotating registers)
Wire length	198,673 um routed (global routing 64.8 % of capacity)
Setup WNS, typical (nom / min / max RC)	**+1.991 / +2.407 / +1.587 ns** at 15.625 ns
Setup WNS, fast	+4.41 ns
Setup WNS, slow (not enforced)	-12.17 ns, 274 endpoints

Metric	Value
Hold WNS	**+0.0096 ns** (min_ff), 0 violations at all 9 corners; +0.16 ns at tt
Max slew	753 pins over 0.75 ns at nom_tt (worst 1.40 ns; 1.49 ns at max_tt); not a flow failure
Max capacitance	0 at tt and ff, 24 pins at ss
Max fanout	1 net with 17 loads (limit 16)
Power (STA estimate, tt, 64 MHz, default activity)	6.6 mW (internal 4.5 mW, switching 2.1 mW, leakage 0.07 uW)
IR drop	0.17 mV worst

Max slew explained. Every tt slew violation comes from one choice of the resizer: it used `sky130_fd_sc_hd__clkdlybuf4s25_1`, a clock-delay cell with weak output drive, for all 94 fanout buffers it inserted, and some of them drive 16 loads. The slow edges are included in the STA delay calculation (the setup slack above already contains them), and the worst tt slew (1.49 ns) is inside the 1.5 ns range for which the sky130_fd_sc_hd timing tables are characterised, so the delays are not extrapolated. They cost some speed and dynamic power, not correctness. Removing them would mean excluding `clkdlybuf4s*` from resizer buffering and re-hardening; this was not done because the 2x2 iteration budget (8 runs) had been used and the design already passes.

8.6 Signoff

Check	Tool	Result
Routing DRC	OpenROAD TritonRoute	0 violations
Layout DRC	Magic 8.3 (sky130A rules)	0
LVS	Netgen	clean: 0 device, net, pin or property differences
Antenna	OpenROAD	0 violating nets / pins (repair rounds: 10 -> 2 -> 1 -> 0 nets)
Illegal overlap	Magic	0
Power grid	OpenROAD PSM	0 violations
Setup (enforced at tt)	OpenSTA, 9 corners	met, +1.587 ns worst
Hold (enforced at all corners)	OpenSTA, 9 corners	met, +0.0096 ns worst
Lint	Verilator	0 errors; 9 harmless warnings: 8 x PINMISSING (power pins of the sky130 cells TinyQV instantiates directly in `counter.v` and `register.v`; the flow connects power), 1 x WIDTHTRUNC (`UART_DIV_RESET` = 555 computed in 32 bits, stored in 13: no bits lost)
Tiny Tapeout precheck (`scripts/precheck.sh`)	Magic 8.3.623, KLayout 0.30.7	**passed**
- Magic DRC		0
- KLayout FEOL, BEOL, off-grid, zero-area, nwell/urpm, pin label purposes		0 items each
- top-cell name, forbidden layers, prBoundary, pins vs. 2x2 template, Verilog syntax		pass
Gate-level simulation	cocotb + Icarus, powered netlist	11/11 pass, cycle-identical to RTL (Section 7.4)

The precheck reports are kept in `reports/precheck/`.

9. FPGA prototyping

`fpga/` maps the unmodified chip RTL onto a PYNQ-Z2 (Zynq xc7z020, PL only):

- MMCM: 125 MHz board clock to 64 MHz;
- `qspi_pmod_emu.v`: a synthesizable, cycle-equivalent version of the QSPI Pmod model in block RAM (3 x 64 KB, nibble-wide), flash preloaded with the firmware. SCK edges are seen one clock late and reads are registered, so data return exactly two clocks later than the simulation model; the chip's read latency is set to 2 during reset;
- `uart_watch.v`: decodes the chip's UART and lights RGB LEDs on `Hello, TT!`, `SELFTEST PASS` or `FAIL`, so a pass is visible without a PC;
- buttons, switches, LEDs and both Pmods mapped to the chip pins (`docs/fpga.md`).

Verification of the FPGA build. Two testbenches run the real firmware: `fpga/sim/tb_fpga.v` bypasses the MMCM and uses the FAST firmware mode, and `fpga/sim/tb_fpga_board.v` takes no shortcut: 125 MHz into the Xilinx UNISIM MMCM model, 115200 baud, and only the board pins are driven and observed (boot text and all self-test lines, status LEDs, BTN1/BTN2 interrupts, UART echo through Pmod JA, GPIO mirror mode with the switches, BTN3 and Pmod JA4; 12 checks). The whole Vivado flow is one script, `fpga/pynq_z2/build_on_windows.ps1`: both RTL simulations in xsim, synthesis, implementation, reports, bitstream, and finally the board test on Vivado's post-route functional netlist, which is the logic inside the bitstream.

Step (Vivado 2025.1, 10 Nov 2026)	Result
xsim RTL, `tb_fpga.v`	pass (same output as Icarus)
xsim RTL, `tb_fpga_board.v`	pass, 12/12 checks; banner 1,134 us and self-test 7,801 us after BTN0
Synthesis	0 errors, 0 critical warnings
Timing at 64 MHz (post-route)	setup WNS +2.126 ns, hold WHS +0.052 ns, 0 failing endpoints
Resources	2,035 LUTs (3.8 %), 952 FFs, 32 BRAM36, 1 MMCM; the chip itself 1,155 LUTs / 661 FFs
xsim post-route netlist, `tb_fpga_board.v`	pass, 12/12 checks, identical UART text and times to the RTL

Two things had to be handled for Vivado that Icarus and Yosys accept silently: `project.v` sets `` `default_nettype none ``, which Vivado carries into the TinyQV files read after it (their ports have no net type), so TinyQV is read first; and `uart_rx.v` has a function without arguments, so the chip RTL is read as SystemVerilog. The flash ROM is built from LUTs (Vivado trims it to the 16 K nibbles the firmware fills); the PSRAMs use 32 block RAMs. Reports and logs are in `reports/fpga/`. The board test itself is scheduled for when the board is available.

10. Discussion

Area. The design uses about 46,000 um2 of standard cells after synthesis (Table A1) and 56,900 um2 after placement, clock-tree synthesis and repair (tap cells, clock buffers, fanout and timing buffers, antenna diodes), i.e. 78 % of the 2x2 core. The register file (13,500 um2 including delay cells) and the CPU control

(11,700 um2) dominate; peripherals are about 8,600 um2. There is no room left in 2x2 for more logic without giving up routability.

Performance. One simple instruction takes at least 8 clocks; fetch from QSPI adds stalls after every branch. With code in flash and stack in PSRAM, a C interrupt round trip costs about 2,400 clocks. The design suits control tasks, not computation.

Limitations. No illegal-instruction trap; no counters (use TIMER_COUNT); a one-byte UART receive buffer; slow-corner timing below 64 MHz (about 36 MHz at ss, 100 C, 1.60 V); slow edges on 94 weak fanout buffers (Section 8.5); the chip depends on the RP2040 (or equivalent) to configure the QSPI devices before reset.

Risks for the shuttle. (1) The Tiny Tapeout GitHub Action re-hardens the design from the repository. Detailed routing needed 20 passes to reach zero violations, so the result depends on the flow being deterministic; a local reproduction run from the committed sources reproduced the layout exactly (Section 12). (2) The QSPI read latency depends on the demo board and must be chosen at reset (SD2:SD0); the firmware banner at the default 115200 baud tells whether it is right.

Future work. Exclude `clkdlybuf4s*` from resizer buffering to remove the slew warnings; a 3x2 version would leave room for a larger UART buffer, the Zicntr counters and the SPI controller; FPGA test on the PYNQ-Z2.

Lessons. (1) For serial datapaths, hold-repair margin is a first-order routability parameter. (2) Repairing setup only at enforced corners avoids futile over-buffering. (3) Below a design-dependent minimum the placement density setting has no effect, and slightly above it can help routing. (4) A firmware self-test that runs unchanged in RTL, gates, FPGA and silicon is the cheapest broad check and found two real core bugs.

---

11. Conclusion

The SoC meets the functional specification and fits the 2x2-tile budget of Tiny Tapeout SKY 26e. After eight documented closure iterations the layout is clean (DRC, LVS, antenna, hold at all corners, setup at the typical corners at 64 MHz) and passes the Tiny Tapeout precheck; the same tests pass on RTL and gate level. All design files, firmware, tests, FPGA harness, scripts, logs and reports are in the repository, and the whole flow reruns with one command. What remains is manual: publishing the repository, checking the GitHub Actions, submitting to the shuttle, and the FPGA board test.

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12. Reproducibility

Tool	Version
Ubuntu (WSL2)	22.04.5
LibreLane	3.0.14 (Docker `ghcr.io/librelane/librelane:3.0.14`)
sky130 PDK	open_pdks 8afc8346a57fe1ab7934ba5a6056ea8b43078e71 (via ciel)
tt-support-tools	main @ 37f5534 (2026-11-04)
cocotb	2.1.0; Icarus Verilog 11.0; Python 3.12 (uv)
RISC-V GCC	xPack riscv-none-elf-gcc 14.3.0-1
Precheck tools	Magic 8.3.623 and KLayout 0.30.7 (from the LibreLane image), klayout Python 0.30.13
FPGA	Vivado 2025.1 (Windows)

```
scripts/check_env.sh          # install / verify everything (no sudo)
scripts/run_all.sh            # firmware, RTL tests, FPGA sim, harden, GL tests, reports
scripts/precheck.sh           # Tiny Tapeout precheck on tt_submission/
scripts/gl_test.sh            # gate-level tests on runs/wokwi/final/pnl
make -C fpga sim              # FPGA harness simulation
./tt/tt_tool.py --pdk sky130A --create-png   # gds_render.png (Figure 1)
scripts/docs/module_map.sh    # docs/images/floorplan_modules.png (Figure 2)
scripts/docs/make_pdf.sh      # docs/pdf/*.pdf
```

The Vivado flow runs from Windows PowerShell (it stages the sources locally):

```
powershell -ExecutionPolicy Bypass -File \\wsl.localhost\Ubuntu-22.04\home\santhosh\tt-sky26d-rvsoc\fpga\pynq_z2\build_on_windows.ps1
```

Run times on the laptop used (Intel i5-1135G7, 4 cores / 8 threads, 7.8 GB RAM; WSL2 with 8 threads and 7 GB): one hardening run 5 h 12 min (most of it detailed routing), RTL tests 44 min, gate-level tests 56 min, precheck about 10 min; Vivado flow: xsim RTL simulations 18 min, synthesis to bitstream 16 min, post-route netlist simulation 50 min. Vivado and a hardening run should not run at the same time on this machine: together they exceed its memory and both slow down several times (synthesis took 25 min instead of 9 when they overlapped).

Determinism check. The final run (it8) was hardened in a scratch copy. A second run was started from the committed sources of the main project (`.work/repro`) to confirm that a clean re-harden, as done by the GitHub Action, gives the same layout. Result (10 Nov 2026): **identical**. The synthesized netlist, the powered netlist and the DEF have the same SHA-256 hashes, KLayout's `LayoutDiff` finds the two GDS files geometrically identical, every metric in `metrics.json` is equal except run times (for example setup +1.5870 ns, hold +0.00963 ns, 10,897 instances, 198,673 um of wire), and all 82 detailed-routing passes had the same violation counts (5,943 -> 3,463 -> ... -> 0 in the first round, then the same sequences in the three rounds after antenna-diode insertion). The reproduction ran with several days of laptop sleep and two manual pauses in the middle, which did not change anything. With the same tool versions, the GitHub Action's re-harden should therefore produce the same layout.

References

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2. Tiny Tapeout, "Hardening Tiny Tapeout projects locally", https://www.tinytapeout.com/guides/local-hardening
3. Tiny Tapeout SKY 26e shuttle repository, https://github.com/TinyTapeout/tinytapeout-sky-26e (SKY 26d: https://github.com/TinyTapeout/tinytapeout-sky-26d)
4. LibreLane, https://github.com/librelane/librelane
5. SkyWater SKY130 PDK, https://github.com/google/skywater-pdk
6. A. Waterman, K. Asanovic (eds.), *The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA*, RISC-V International.
7. B. Marshall, "uart" (MIT), as used in TinyQV.
8. mole99, "QSPI Pmod", https://github.com/mole99/qspi-pmod

Appendix A. Area by block (final RTL, sky130 tt library, before placement)

Block	Area (um2)
tinyqv_registers (incl. 364 delay cells)	13,505
tinyqv_cpu (fetch, decode registers, sequencing)	11,720
tinyqv_core (ALU path, CSRs, traps)	4,725
qspi_controller	3,053
tinyqv_mem_ctrl	2,544
rvsoc_timer	2,332
tinyqv_decoder	1,779
tt_um_santhosh_rvsoc (peripheral decode, GPIO, UART_DIV)	1,607
tinyqv_shifter	1,485
uart_rx	1,399
uart_tx	1,251
tinyQV wrapper, ALU	801
Total	**about 46,200**

Appendix B. Repository map

Path	Contents
`src/`	RTL; `project.v` top, `timer.v`, `tinyQV/` (core, UART, licences, `UPSTREAM.txt`)
`firmware/`	`start.S`, `isa_test.S`, `main.c`, `rvsoc.h`, `link.ld`, `Makefile`
`test/`	`tb.v`, `sim_qspi.v`, `test_soc.py`, `test_isa_random.py`, `test_util.py`, `rvsoc_fw.hex`
`fpga/`	`rtl/qspi_pmod_emu.v`, `rtl/uart_watch.v`, `pynq_z2/` (top, XDC, Vivado scripts), `sim/` (`tb_fpga.v`, `tb_fpga_board.v`), `tools/hex2nib.py`
`scripts/`	`check_env.sh`, `env.sh`, `run_all.sh`, `harden.sh`, `gl_test.sh`, `precheck.sh`, `sweep_tiles.sh`, `set_gh_user.sh`, `metrics.py`, `make_summary.py`, `live.sh`; `docs/` (module map)
`docs/`	`USER_GUIDE.md` (user and application guide), `info.md` (TT datasheet), `IO_GUIDE.md`, `fpga.md`, this report, `images/` (layout renders, module map), `pdf/` (PDF versions)
`reports/`	`summary.md`, `sweep.md`, `precheck/` (TT precheck reports), `fpga/` (Vivado reports, xsim logs, `vivado_summary.txt`)
`runs/` (not in git)	`wokwi/` = final LibreLane run (GDS, LEF, netlists, SPEF, SDF, STA reports); `iterations/` = metrics of each run
`tt_submission/` (not in git)	the files Tiny Tapeout takes from the run: GDS, OAS, LEF, Verilog, SPEF
`logs/` (not in git)	every tool log: harden, tests, precheck, environment
`STATUS.md`	phase log, decisions, remaining manual steps